

→ OSI model :-

1. Why was the OSI model created?

⇒ In the early days of computers networking, different manufacturers such as **IBM, Xerox, and Apple** developed their own networking technologies, hardware, software, and communication protocols.

Since, there was **no common standard**, computers made by different companies could not communicate properly with each other.

To solve this problem, the **OSI** (Open Systems Interconnections) model was developed as a **standard reference model** for communication b/w computer systems.

Problems before the OSI model :-

1. **No Standard Communication** :-

Each company used its own communication rules (protocols). As a result, computers from **different manufacturers** could not exchange data easily.

2. **Compatibility problem** :-

Different computer systems were incompatible with each other.

Eg:

IBM computer could not communicate properly with an Apple computer because both used different communication protocols.

Protocol :- A set of rules used for communication b/w devices.

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3. Different Hardware :-

Different manufactures used different :

- Network Cables
- Connectors
- Communication devices.

This made networking difficult.

4. Different software :-

Each company developed its own networking software.

Since, there was no common standard, communication b/w different systems become difficult.

5. High cost :-

Every company had to develop its own networking technology.

This increase the cost of development, maintenance and communication.

6. Lack of interoperability :-

Interoperability means the ability of different computer systems or devices to communicate and work together successfully.

→ Need for the OSI model :-

A common networking standard was required so that :

- Different computer system could communicate with each other.
- Different manufacturers could follow the same communication rules.
- Network design become easier.
- Future networking technologies could be developed more easily.

→ Why was layered introduced?

Instead of performing all networking functions together, the OSI model divided the communication process into **seven different layers**.

Each layer performs a specific task and provide services to the layer above it :

This layered approach makes networking :

- Easier to understand
- Easier to design
- Easier to maintain

~~Simple~~

→ Objectives of the OSI Model :-

The main objectives of the OSI model are :

- To establish a standard for network communication.
- To enable communication b/w different-computer systems.
- To improve interoperability.
- To divide networking into separate functional layers.
- To reduce development complexities.
- To allow independent development of each networking layer.

- Advantages of creating the OSI model :-
- Standardized communication & Establishing common rules for communication.
 - Better compatibility
 - Easier network design
 - Modular architecture
 - Independent layer development.

• Eg;

Suppose a Samsung phone sends a whatsapp message to an iPhone.

Even though both devices are manufactured by different companies, they can communicate successfully because modern networking follows standardized communication protocols.

The OSI model explains how such communication should take place.

2. What is the OSI model?

OSI (Open System Interconnections) model is a **Seven-layer reference model** that describes how data is transmitted from one computer to another over a network.

It is developed by the **International Organization for Standardization (ISO)** to standardize communication b/w different computer systems in a network.

O → open → Open to all manufacturers and vendors, Any company can follow this standard.

S → Systems → Computers, laptops, mobile phones, servers, printers etc.

I → Interconnections → Connection different systems so they can exchange data.

Open System Interconnections means connecting different computer systems using a common standard.

→ Who developed the OSI model?

The OSI model was developed by the:

International Organization for Standardization (ISO)

It was introduced in **1984**.

→ Why it is called a "Reference Model"?

The OSI model is called a **Reference model** because it serves as a **guide** or **blueprint** for designing and understanding computer network.

It does **not** define a complete communication protocol.

instead, it explain:

- What function should be performed during communication.
- Which layer is responsible for each function.
- How different layers interacts with each other.

→ What does the OSI model do?

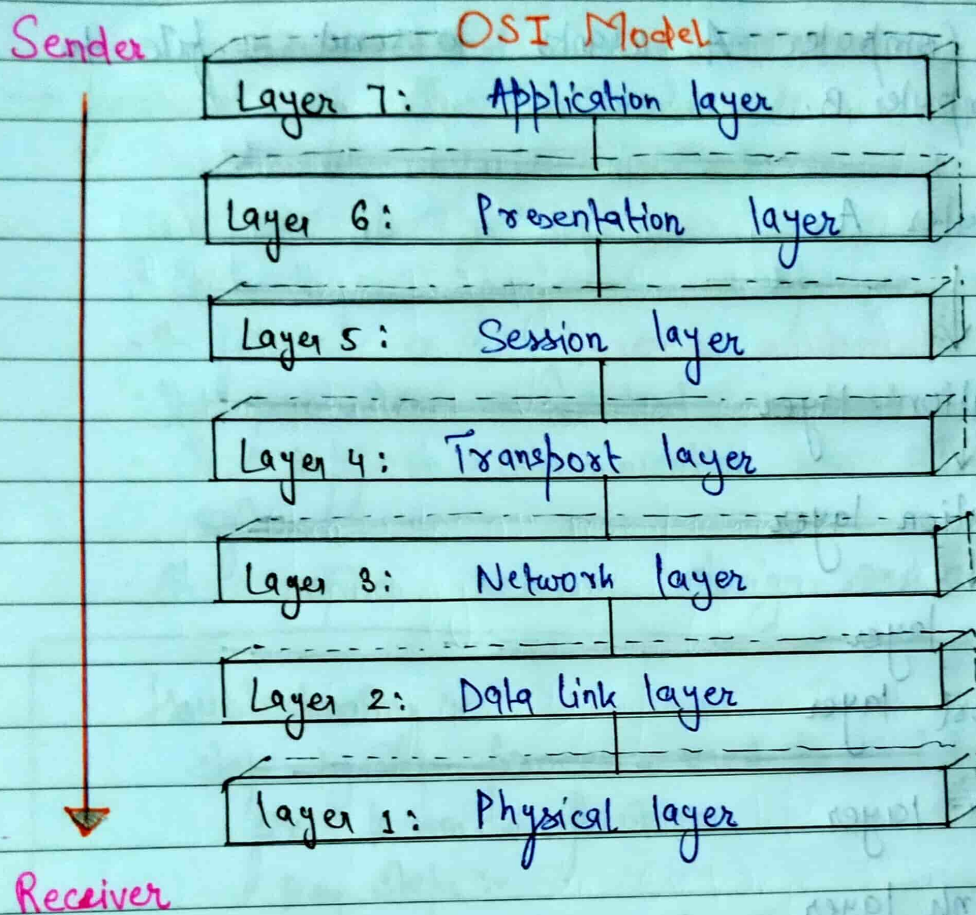
The OSI model divides the communication process into **seven different layers**.

Each layer performs a special tasks and passes the ~~process~~ processes data to the next layer.

This makes communication:

- Organized
- Standardized
- Easier to understand

→ Seven layers of the OSI model is



Each layer has a specific responsibility.

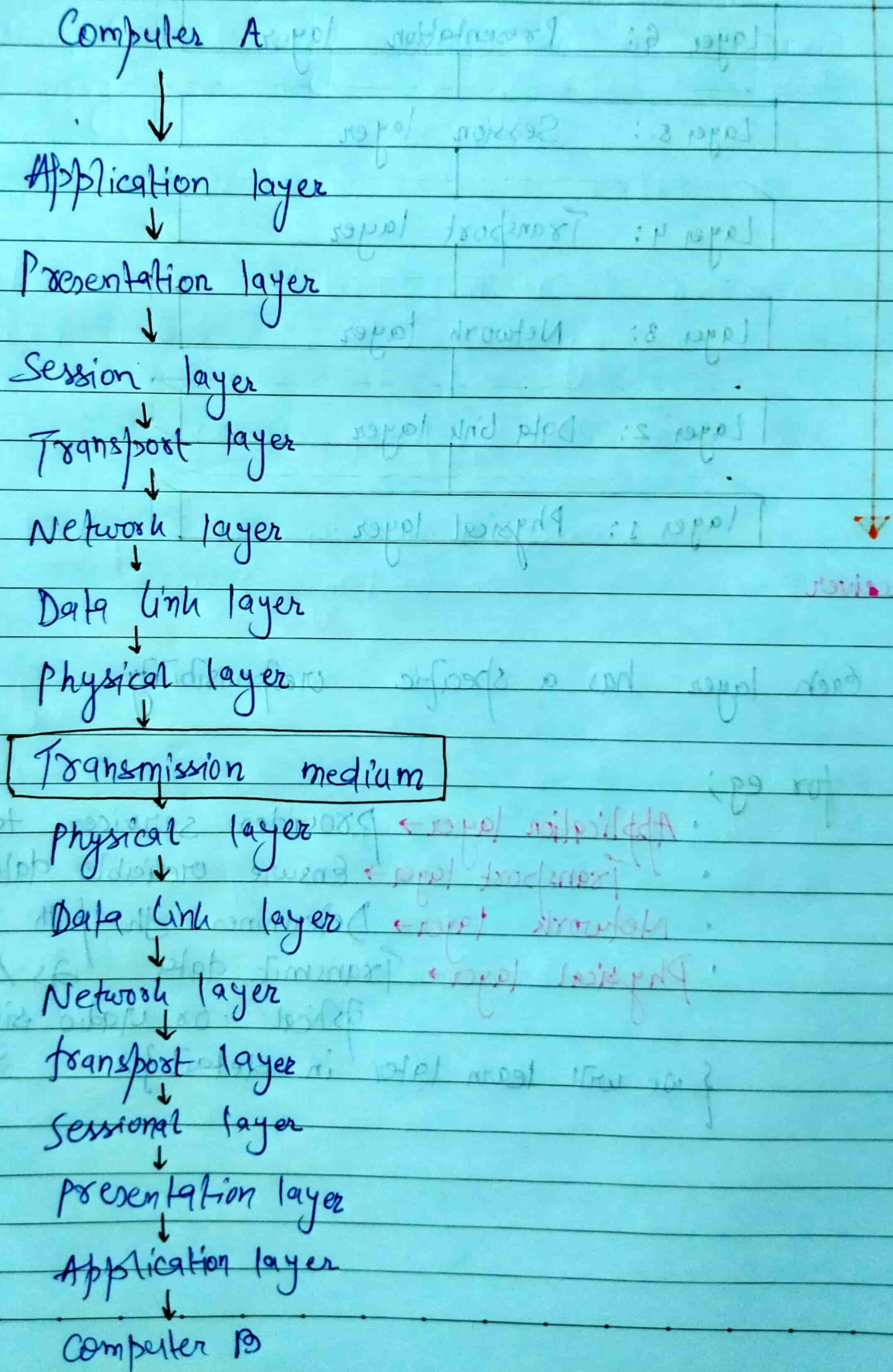
for eg;

- Application layer → provides services to users.
- Transport layer → Ensure reliable data delivery.
- Network layer → Determines the path for data.
- Physical layer → Transmit data as electrical, optical, or radio signal.

{ we will learn later in detail }.

→ Working of the OSI Model is

Suppose Computer A wants to send a file to Computer B.



- On the **sender side**, data moves from **layer 7 to layer 1**.
- On the **receiver side**, data moves from **layer 1 to layer 7**.

→ features of the OSI model:~

- It consists of **7 layers**.
- Each layer performs a specific function.
- It follows a **layered architecture**.
- It provides a standard method for communication.
- It supports communication b/w different computer systems.
- It simplifies network design, ~~and~~ and troubleshooting.

Troubleshooting is

step-by-step process used to identify, diagnose and fix technical problem.

Key steps:~

- Identify the problems
- Gather clues
- Test ideas
- Fix and verify

Common quick fixes:~

- Restart the system: clears temporary glitch glitches.

→ Why is the OSI model important?

The OSI model helps us to:

- Understand how computer network work.
- Design communication systems systematically.
- Identify network problem layer by layer.
- learn networking concepts in an organized way.

3.4 Layer 1: Physical layer :-

The physical layer is the **first (lowest)** layer of the OSI Model.

It is responsible for the **physical transmission of raw bits (0s and 1s)** from one device to another a communication medium.

It deals with the **electrical, optical, or wireless signals** used to transmit data.

In simple terms,

The physical layer is the first layer of the OSI model that is responsible for transmitting raw bits (0s and 1s) over a physical communication medium using electrical, optical, or radio signals.

Layer 1: Physical layer



Transmission medium
(Cable/Fibre/Air)

The physical layer is directly connected to the communication medium.

→ Why is the physical layer needed?
Computer understand only **Binary data (0 and 1)**.
However, cables, optical fibre, and wireless channels **cannot transmit 0s and 1s directly**.

∴ the physical layer converts bits into ~~bits~~ **signals** that can ~~active~~ travel through the medium.

Eg;

Computer A

Data:

(10110010)



Physical layer



Electrical signal



cables



Receiver



Physical layer



10110010



Computer B

Bit :- A bit (Binary Digit) is the smallest unit of data in a computer (0,1).

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→ Data Unit (PDU) :- Every OSI layer has its own **protocol data unit (PDU)**.

Layer

PDU

Application

Data

Presentation

Data

Session

Data

Transport

Segment

Network

Packet

Data link

Frame

Physical

Bits

→ Function of Physical layer :-

1. **Transmission of raw bits** :-

The physical layer sends raw binary bits from the sender to the receiver.

Eg: 1011001

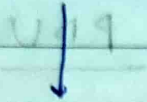
2. **Signal Conversion** :-

Computers generate bits.

Communication media carry signals.

The physical layer converts :

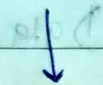
Bits



Electrical signals

or

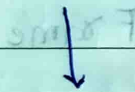
Bits



Light signals

or

Bits



Radio signals

3.4 Transmission medium :-

The physical layer specifies the medium through which data travels.

- Eg ;
- Coaxial cables
 - Optical fibre
 - Radio waves
 - Microwave
 - Infrared

{ we ~~have~~ have already studied this }

4.7 Data rate :-

The physical layer defines the transmission speed.

Eg;

- 100 Mbps
- 1 Gbps
- 10 Gbps

5.7 Physical topology :-

It defines how devices are physically connected.

Eg;

- BUS
- Star
- Ring
- Mesh
- Tree

} we have already studied this }

6.7 Transmission mode :-

Defined direction of communication.

- Simplex
- Half duplex
- Full duplex

} we have already studied this }

→ Physical layer Devices :-
Devices that operate at layer 1:

1. Hub
2. Repeater
3. Modem
4. Cables
5. Connectors
6. Network Interface cards (physical layer functions)

1. HUB :-

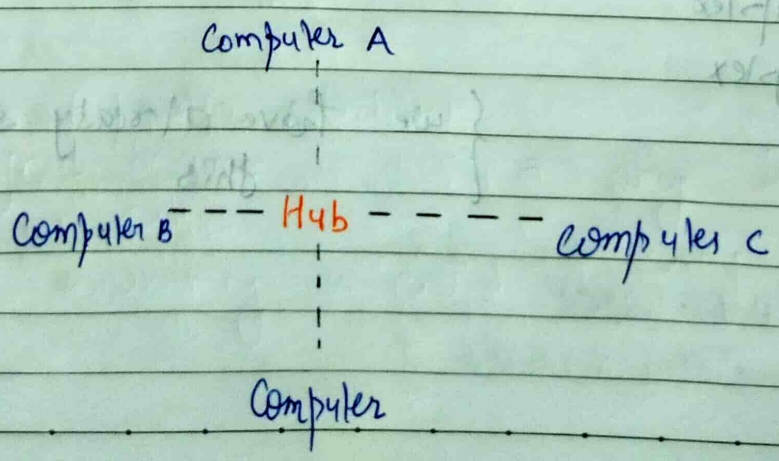
A hub is a simple networking that connects multiple computers in a LAN and broadcast the received data to all connected devices.

In other words;

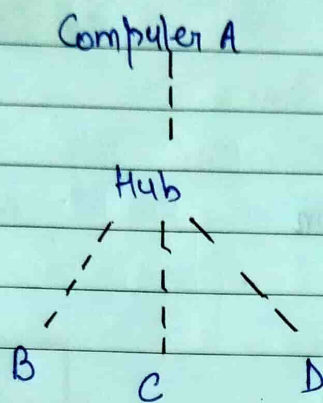
A Hub is a physical layer networking device that connects multiple computer and forwards incoming bits to all its port without checking the destination address.

Working :-

Suppose there are four computer connected to a HUB



here, Computer A sends data to Computer C. through the hub.



Instead of sending only to computer C, the Hub sends it to:

- Computer B
- Computer C
- Computer D

→ All computer receive the data.

- Only the intended receiver processes it.
- The remaining computers simply ignore it.

→ Why does Hub Broadcast?

The Hub works only with electrical signals. It cannot read:

- IP Address
- MAC Address
- Frame
- Packet

∴, it does not know where the data should go.

The easiest solution is:

"Sends the data to everyone".

Collision: means that two or more computers try to send data at the same time, causing the data signals to interfere with each other.

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→ Characteristics of Hub:~

- Physical layer device
- No MAC Address table
- No IP Address knowledge
- Broadcast data
- Half duplex communication
- Low cost
- Low security

→ Advantages:~

- Cheap
- Easy to install
- Connects multiple computers

→ Disadvantages:~

- Broadcast every message
- High network traffic
- More collision
- Low security
- Slow performance.

2.) Repeater:~

A Repeater is a physical layer device that receives weak signals, regenerates (strengthens) them, and transmits them.

→ Why is Repeater Needed?

Signal become weaker as they travel through a cable.

This weakening is called attenuation.

Strong signal



long cable



Weak signal

If the signal become too weak, the receiver cannot understand it.

→ Working :-

Computer



Weak Signal



Repeater



Strong Signal



Receiver

The repeater regenerates the signal and sends it again.

→ Functions :-

- Receives weak signals
- Regenerates signals
- Extends communication distance

→ Advantages :-

- Extends network ranges
- Improves signal quality
- Low cost

→ Disadvantages :-

- Cannot filter traffic
- Cannot understand address
- Cannot make routing decisions

3.Y Modem :-

A MODEM is a device that converts digital signal into analog signal and analog signal back into digital signals.

The word **MODEM** comes from :-

MO - Modulator

DEM - Demodulator

→ Why is Modem needed?

Computers understand digital signals (0 and 1).
Traditional telephone lines carry analog signals.
A modem converts b/w these two signal types.

→ Working of a Modem

Computer
Digital signal



Modem



Analog signal



Telephone line



Modem



Digital signal



Computer

→ Functions:

- Digital to Analog conversion (Modulation)
- Analog to Digital conversion (Demodulation)

→ Advantages:

- Enables internet over telephone lines
- Converts signal formats
- Supports long distance communication

→ Disadvantages:

- Traditional-dialup modems are slow compared to modern broadband technologies.

4. Network Cables :-

A cable is the **physical medium** through which signals travel.

→ Types of cables :-

(a) Twisted pair cables

Used in :-

- LAN
- Ethernet

→ Advantages :-

- Cheap

→ Disadvantages :-

- Short distance

(b) Coaxial cable :-

Used in :-

- Cable TV
- Internet

→ Advantages :-

- Better shielding
- longer distance than twisted pair

(c) Optical fibre :-

Uses light instead of electricity.

→ Advantages :-

- Very high speed
- Very long distance

→ Disadvantages :-

- Costly
- Installation is more difficult.

} We have already studied this

5. Connectors :-

Connectors join network devices and cables.

(i) → RJ45 Connectors

Used with :

- Ethernet Cable
- LAN

Cable ===== RJ45 ===== Switch

(ii) fibre Connectors :-

Used with :

• optical fibre

Eg:

- SC
- ST
- LC

Q. Network Interface Card (NIC):

A Network Interface Card (NIC) allows a computer to **connect** to a network.

Although a NIC also performs **Data link layer functions** (such as using a MAC address), it has a **physical layer responsibility** like transmitting and receiving signals.

functions:

- Connects computers to network.
- Sends signal.
- Receive signal.
- Provides a physical interface.

Ans